

Math 2270 Review for Quiz 4 part B

- ① Define $V^\perp$ and compute $V^\perp$ where $V = \left\langle \begin{pmatrix} 2 \\ 1 \\ 0 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \right\rangle$
- ② Show $(V^\perp)^\perp = V$

(Hint: show $V \subset (V^\perp)^\perp$ and compute dimensions)

- ③ Show that $V^\perp \cap V = \{0\}$ and $V + V^\perp = \mathbb{R}^n$

- ④ Project $\begin{pmatrix} 1 \\ 2 \\ 3 \\ 1 \\ 2 \\ 3 \end{pmatrix}$ on to $\left\langle \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \\ 1 \\ 0 \\ -1 \end{pmatrix} \right\rangle$ You may use the formula $P = A(A^T A)^{-1} A^T$

- ⑤ Project $\begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix}$ on to $\left\langle \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} \right\rangle$

- ⑥ Let $P = A(A^T A)^{-1} A^T$. Show that $P^2 = P$, $P^T = P$

- ⑦ Prove that $N(A^T A) = N(A)$ and deduce that if A has max rank then $A^T A$ is invertible

(Here $A \in M_{m,n}$ with $m \geq n$)

- ⑧ Find the line of best fit for $\begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \end{pmatrix}, \begin{pmatrix} -1 \\ -1 \end{pmatrix}$.
(Then find the parabola of best fit).